

## Defining Each Type of Operator in Bash

Definition:

Operators in Bash are special symbols or keywords used to perform operations such as arithmetic calculations, comparisons, logic, string testing, and file handling.

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### 1. Arithmetic Operators

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Used for mathematical calculations.

Syntax:

```
$(( expression ))
```

Example:

```
a=10
```

```
b=5
```

```
echo $((a + b)) # 15
```

```
echo $((a ** b)) # 100000
```

Operators:

+ Addition

- Subtraction

\* Multiplication

/ Division

% Modulus

\*\* Exponentiation

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### 2. Relational (Comparison) Operators

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Used to compare two numeric values.

Syntax:

```
[ expression ]
```

Example:

```
a=8
b=10
if [ $a -lt $b ]; then
    echo "a is less than b"
fi
```

Operators:

- eq Equal to
- ne Not equal to
- gt Greater than
- lt Less than
- ge Greater or equal
- le Less or equal

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### 3. Boolean (Logical) Operators

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Used to combine or invert conditions.

Syntax:

```
[[ condition1 && condition2 ]]
```

Example:

```
a=5
b=15
if [[ $a -lt 10 && $b -gt 10 ]]; then
    echo "Both conditions true"
```

fi

Operators:

! NOT

&& AND

|| OR

---

#### 4. String Operators

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Used to compare or test strings.

Example:

```
str1="hello"
```

```
str2="world"
```

```
if [ "$str1" != "$str2" ]; then
```

```
    echo "Strings differ"
```

```
fi
```

Operators:

= Equal

!= Not equal

-z String empty

-n String not empty

< Less than alphabetically

> Greater than alphabetically

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#### 5. File Test Operators

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Used to check file existence, type, and permissions.

Example:

```
file="/etc/passwd"
if [ -f "$file" ]; then
    echo "File exists"
fi
```

Operators:

- e File exists
- f Regular file
- d Directory
- r Readable
- w Writable
- x Executable
- s Not empty

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## 6. Assignment Operators

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Used to assign or modify variable values.

Example:

```
a=5
((a+=3)) # a = 8
((a*=2)) # a = 16
```

Operators:

- = Assign
- += Add and assign
- = Subtract and assign
- \*= Multiply and assign
- /= Divide and assign

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Summary Table

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| Type       | Purpose            | Example                       |  |
|------------|--------------------|-------------------------------|--|
| -----      | -----              | -----                         |  |
| Arithmetic | Math operations    | \$((a + b))                   |  |
| Relational | Number comparison  | [ \$a -lt \$b ]               |  |
| Logical    | Combine conditions | [[ \$a -gt 0 && \$b -lt 10 ]] |  |
| String     | Compare text       | [ "\$a" = "\$b" ]             |  |
| File Test  | File checking      | [ -f /etc/passwd ]            |  |
| Assignment | Assign/update vars | ((a+=1))                      |  |